Analysis of Internet Traffic Through Major US Backbone Links

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ABSTRACT

Internet measurement and analysis of high-traffic backbone links across the world give us key insight into how the Internet is used in real-world scenarios. Knowledge of usage, changes in use by location, and port usage allows network providers and designers of protocols to make changes to networks and applications as needed, and to make predictions about future network traffic. We study behaviors of Internet traffic across three different links distributed across the US, and how Internet usage has changed over time and can differ throughout the regions that these links are located in. Specifically, we analyze international and domestic traffic across these links, the change in use of secure protocols over time, and other general measurements such as the adoption of IPv6 over time. This work has given us a deeper understanding of networking research, and in particular the challenges that arrive with aggregating the data, understanding how the data can be used, and applying this to interesting questions about the Internet.

CCS CONCEPTS

• **Networks** → **Network measurement**; **Public Internet**.

1 INTRODUCTION

Aggregating data from busy links is generally difficult to carry out, since it requires permission to access and monitor backbone links, high-speed monitors to collect information, and in some cases, post-processing for public release. In addition, choosing how to analyze and make conclusions about Internet traffic is a difficult task, often exacerbated by the large amounts of data and processing power necessary. We chose this project because we were interested in how large-scale analysis on Internet links were conducted, what data was important to extract from a dataset of this sort, and how we could use this data to analyze properties about the specific links, how different links may compare and contrast based on location, and how Internet usage changes over time.

While the data for the three backbone links we discuss in this paper is publicly available, we have not seen much research using this dataset that focuses on our interests of tracking the network behavior across location and over time. Thus, we believe this is one of the first high-level look at the CAIDA dataset presenting information on traffic usage based on country identification, overall protocol usage, emphasizing the emergence of secure protocols such as HTTPS.

In this paper, we use publicly available Internet traces dataset to evaluate Internet traffic at three distinct backbone links across the US and report results about them. For each of these datasets, we extract summary statistics about IP address and their associated IP prefix frequencies along with the port usage. From this data, we evaluate trends in Internet usage and traffic related to the amount of traffic, geological information based on the traffic, and the port usage throughout the links and how each changes over time.

From our analysis, we found interesting results about the Internet usage from these three links. Firstly, we found a general upward trend in the use of secure protocols over time. In particular, we see an increase in the ratio of secure protocols in widely used applications vs. unsecure protocols, specifically HTTPS and secure email protocols, and general stagnation in the adoption of DNS over TLS. We also analyzed the relationship between international and domestic traffic for the provided links. Within our analysis, we found at most a 5% difference between domestic and international traffic between the Chicago and San Jose links in 2013. We also found no discernable trends between domestic and international traffic through time from our analysis Chicago traffic over time. The largest change we saw was with multicast, as its usage went from 25% of source and 44% of destination packets for 2008 to less than 1% for both source and destination in 2016.

The outline of the paper is as follows: In section 2, we begin with background information and related works from which we took inspiration. In section 3, we give a high level overview of the data sources used. This is followed by a detailed description of our data processing methods in section 4, along with a high-level overview of results gained from this data processing. In section 5, we analyze Internet traffic in the backbone links based on its origin and destination, giving detailed analysis on geolocation information of the IPs used across these links. In section 6, we check privileged port usage across the backbone links, and analyze the adoption of secure protocols over time. Finally, we present our reflections from this project in section 7, and conclude with section 8.

2 BACKGROUND AND RELATED WORKS

Internet measurement has been an emerging research topic as tools and processing power develop to allow us to analyze large sets of Internet traffic for the last few decades. A seminal paper about Internet measurement is by Paxson, which talks about Internet measurement strategies for research projects, such as processing or large datasets, and reproducibility of analysis [11]. Roy et. al. also discusses the difficulty of collecting traffic from high-bandwidth links, specifically in Facebook's datacenters [12]. While the two previous papers take the perspective of analyzing large network or service operators, there has also been research into smaller systems. We took inspiration from the project by Ukani et. al., which details measurement and analysis techniques for analyzing Internet traffic over a University network during the COVID-19 pandemic [15]. We also note papers that also use the CAIDA anonymized dataset, such as [10], which analyzes the use of Port 0 on the traffic from this dataset.

3 DATA SOURCES

3.1 Data sources

We make use of the CAIDA anonymized Internet traces dataset [5] as a source of Internet traffic data. This dataset contains traces of IP packets collected from 3 high speed monitors on major commercial backbone links in the United States. Packets are grouped by backbone link, date collected (year, month), and traffic direction (direction A or B). A summary of the dataset as provided by CAIDA is detailed in Table 1.

Specifically, we use the following datasets:

- equinix-chicago: 4/3/2008, 6/20/2013, 4/6/2016
- equinix-sanjose: 7/17/2008, 6/20/2013
- equinix-nyc: 1/17/2019

We also collected data from equinix-nyc on 3/15/2018, however, we only got through 83% of the data available due to time restrictions, so we will mention this only in section 6.

Each anonymized trace dataset (grouped by date, backbone link, and direction) consists of packets collected in a 1-hour timespan, stored in the pcap format, and readable with software such as tcpdump and Wireshark. Each packet mainly contains the time of sending, IP version, the anonymized source IP address and port, and the anonymized destination IP address and port.

In addition, CAIDA also provides trace statistics for the dataset [6]. We use this information as a sanity check for our measurements.

To determine prefixes of IP addresses, we make use of the CAIDA Routeviews Prefix to AS mappings Dataset (pfx2as) for IPv4 and IPv6 dataset [7]. This dataset provides the mapping from prefixes (both IPv4 and IPv6) to their corresponding autonomous system (AS) number. Data is collected daily since 2005/5/9, and we make use of the data collected on 4/3/2008, 7/17/2008, 6/20/2013, 4/6/2016, and 1/17/2019, which are the same days as the traffic traces were collected.

In addition, we also wish to find AS information such as ownership and geolocation, to perform our analysis. However, since we do not know if there exists any historical AS information dataset, we make use of an AS information dataset from today to help analyze traffic data. Specifically, we use the dataset aggregated by Geoff Huston [9]. This dataset provides the AS name and country of origin of all AS, as of 3/13/2022.

4 DATA PROCESSING

In this section, we present how we processed the datasets, and present a high-level summary of the processed data.

4.1 Processing method

We use tcpdump to decode all pcap files, then pipe the output through a Python script that processes each packet header. For each decoded output, we attempt to read the source and destination IP addresses and port, then perform longest prefix matching to group them into prefixes. We also read the packet type (IP or IP6) to classify the address type expected, though for packets labeled IP6, we also check if it uses an IPv4 address. If an output doesn't contain any IP addresses, we ignore it. During processing, we also encountered outputs in the decoded pcap files with the message "truncated-ip - xx bytes missing!" These message are typically followed by a valid header, specifically source and destination IP addresses and ports. We are unsure if the truncation is caused by the packet anonymization process, or because of link error during original transmission. In our processing pipeline, we make an attempt to detect instances where packets with these errors are followed by valid header details, and we ignore the packet otherwise.

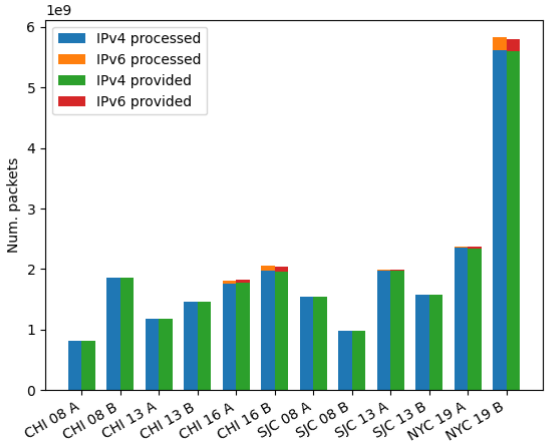


Figure 1: Number of packets traced per dataset

For each dataset (grouped by date, backbone, and location), we count packets grouped by the following categories:

- Source prefixes
- Source IP addresses with and without identified prefixes
- Source privileged ports usage by prefixes
- Destination prefixes
- Destination IP addresses with and without identified prefixes
- Destination privileged ports usage by prefixes
- Source prefix to destination prefix path

We also count the overall number of IPv4 and IPv6 packets, the number of packets without an identified source and destination prefix, and the number of IPv6 packets that use backwards compatible IPv4 addresses.

To identify a prefix of an IP address, we perform longest prefix matching, using the prefixes from the pfx2as dataset on the date of the corresponding dataset. In addition, we also prefix match with the multicast prefix (224.0.0.0/4 for IPv4 and FF00::/8 for IPv6). If an IP address does not match with any prefix, we consider that IP address to not have an identified prefix.

4.2 Summary of data

We present and discuss some high-level statistics of the data observed, which we look at more in-depth in later sections.

Figure 1 shows the number of IPv4 and IPv6 packets that we processed compared to the number of packets reported by CAIDA. As a sanity check, we see that the number of packets processed are mostly the same, outside of a few instances where there is a small observable difference in the plot. In cases were we processed more packets, we believe that it might be because of the truncated

Table 1: Backbone link information

Monitor	equinix-chicago	equinix-sanjose	equinix-nyc
Location	Equinix datacenter in Chicago, IL	Equinix datacenter in San Jose, CA	Equinix datacenter in New York, NY
Backbone link	Tier1 ISP backbone link between Chicago, IL and Seattle, WA	Tier1 ISP backbone link between San Jose, CA and Los Angeles, CA	Backbone link between New York, NY and Sao Paulo. BR
Link speed	Before 12/2010: 9953 Mbps 12/2010 to end of 3/2015: 10 Gbps Since 4/2015: 100 Gbps	Before 6/2010: 9953 Mbps 6/2010 to end of 9/2014: 10 Gbps Since 10/2014: 100 Gbps	Unknown
Direction A	Seattle, WA to Chicago, IL	Los Angeles, CA to San Jose, CA	Sao Paulo, BR to New York, NY
Direction B	Chicago, IL to Seattle, WA	San Jose, CA to Los Angeles, CA	New York, NY to Sao Paulo, BR
Dates collected	4/2008 to 3/2015	7/2008 to 9/2014	3/2018 to 1/2019

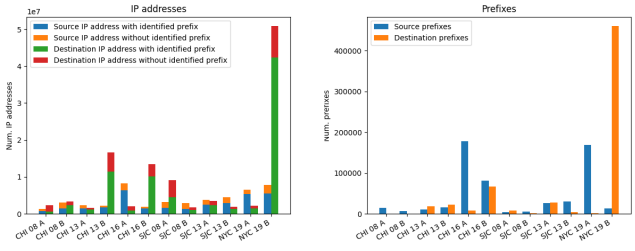


Figure 2: Number of source and destination IP addresses and prefixes observed per dataset

packets that we counted, which might not have been counted by CAIDA. In cases where CAIDA counted more packets, we believe it might have been packets that we ignored by a mistake, possibly by a tcpdump formatting that we ignored. We note that CAIDA also reports IPv6 packets in 2008, before the deployment of IPv6, so there might have been packets that didn't make it into the dataset as well.

We observe that for Chicago and New York, there are more packets traversing from the city (direction B) than there are packets to the city (direction A). For San Jose, there are more packets traversing to the city than there are packets from the city. For Chicago, we think that since it is in the Midwest, it might be responsible for moving traffic between the coasts. The data is generally collected in the early morning, so on the East coast, people would have started working, whereas in the West coast, people are still waking up. This means that Internet usage from the East coast is higher than the West coast at the time of the trace, so we observe more traffic to the West coast than from the West coast. For San Jose, we think that since Los Angeles is the endpoint of many undersea cables, it is likely that it will see more traffic which is then sent to San Jose, then there are packets from San Jose to Los Angeles. For New York, we think that this is because New York is a much larger aggregator of traffic compared to Sao Paulo. In fact, traffic from New York alone more than doubles traffic seen all other datasets. Traffic from New York might have come from the US, as well as from undersea cables from Europe. We will discuss geographic distribution of traffic more in section 5.

Figure 2 shows the number of IP addresses seen in all datasets, separated into source and destination. The number of IP addresses are split between those that matches some prefix in the pfx2as database, and those that don't. It also shows the number of prefixes seen in all datasets, sparated into source and destination.

We note that the prefix data is inaccurate, as we performed prefix matching of addresses in an anonymized address space using prefixes in the real address space, so the prefix groupings doesn't match actual prefix distribution.

Table 2 and Table 3 shows the top 3 source and destination ports of all packets seen in each dataset. We observe that the top are generally HTTP, HTTPS, DNS, or SMTP. We will explore patterns of protocol usage based on this dataset more in section 6.

5 TRAFFIC ORIGIN AND DESTINATION

5.1 Domestic and international traffic

To analyze the amount of domestic vs. international traffic, we have focused on two particular points of view: how the geographic location of traffic changes through time and how it changes based on location. To avoid time-dependent issues between datasets, we have chosen all datasets within this section to be measured from 5:00 AM to 6:00 AM local time. Because we have chosen local time, there is a 2 hour difference between the measurements taken at the San Jose and Chicago links. We would like to mention that the amount of foreign vs. local traffic is heavily dependent on the time of day that a measurement takes place, as the amount of traffic within a particular location typically follows a diurnal pattern [4]. We believe that 6:00 AM is earlier than the typical time an American begins their day. Therefore we expect a majority of the traffic to be foreign for all of these datasets we analyze, and that if the measurement was taken later in the day, we would expect this to flip. We will only be able to analyze the early time.

We expect there to be about 50% traffic from the US and 50% traffic internationally, with multicast being negligable in the calculation for all three years. As time progresses, access to the Internet becomes easier around the world, as Internet services becomes cheaper to provide. In addition, while many major companies such as Google originate in the US, their datacenters are spread across the world that store and cache data help to the load off of links in the States. Thus, as traffic increases outside of the States, we don't expect the traffic to increase any more than the proportion of US traffic.

We also hypothesize that the link between Los Angeles and San Jose link have a considerably larger proportion of foreign traffic when compared to the link connecting Seattle and Chicago. Los

Table 2: Top 3 source ports for each dataset, and percentage of total packets seen in that dataset

	CHI 08 A	CHI 08 B	CHI 13 A	CHI 13 B	CHI 16 A	CHI 16 B
1	80 (26.4%)	80 (37.7%)	80 (46.5%)	80 (49.0%)	80 (11.4%)	80 (25.5%)
2	443 (1.4%)	443 (0.8%)	443 (9.1%)	443 (4.2%)	443 (6.6%)	443 (24.3%)
3	25 (1.2%)	119 (0.8%)	53 (0.3%)	53 (0.5%)	53 (0.2%)	81 (2.6%)

	SJC 08 A	SJC 08 B	SJC 13 A	SJC 13 B	NYC 19 A	NYC 19 B
1	80 (31.7%)	80 (24.2%)	80 (44.7%)	80 (38.8.9%)	443 (40.6%)	443 (4.0%)
2	443 (0.9%)	443 (2.7%)	443 (4.8%)	443 (9.8%)	80 (18.8%)	80 (2.1%)
3	25 (0.7%)	25 (0.9%)	53 (0.4%)	53 (0.5%)	53 (2.0%)	53 (0.8%)

Table 3: Top 3 destination ports for each dataset, and percentage of total packets seen in that dataset

	CHI 08 A	CHI 08 B	CHI 13 A	CHI 13 B	CHI 16 A	CHI 16 B
1	80 (27.3%)	80 (18.1%)	80 (20.0%)	80 (10.2%)	443 (32.8%)	80 (8.9%)
2	53 (0.8%)	25 (1.0%)	443 (3.2%)	443 (3.2%)	80 (19.8%)	443 (5.1%)
3	443 (0.6%)	554 (0.8%)	25 (2.3%)	53 (0.4%)	927 (3.3%)	53 (0.3%)

	SJC 08 A	SJC 08 B	SJC 13 A	SJC 13 B	NYC 19 A	NYC 19 B
1	80 (26.8%)	80 (38.6%)	80 (21.7%)	80 (27.9%)	80 (2.6%)	443 (34. 3%)
2	443 (2.9%)	53 (1.1%)	443 (8.0%)	443 (5.2%)	443 (1.0%)	80 (11.8%)
3	25 (1.6%)	25 (0.9%)	53 (0.4%)	25 (0.8%)	25 (0.9%)	53 (2.3%)

Angeles has a larger amount of submarine cables that connect foreign countries to the US than Seattle does based on the underwater cable map [14], and so we believe much of the traffic, specifically from Asian countries, will likely cross through Los Angeles, which could end up going through our link to San Jose.

For each dataset described, we analyzed the location of the AS that owns each particular prefix. For this description, we will focus on Chicago 2008 source prefixes, but this methodology was applied to both source and destination IP's for each dataset discussed. For each source IP prefix in the Chicago 2008 dataset, we mapped the prefix to it's corresponding AS as mapped by the CAIDA pfx2as dataset. This gave us the amount of source IPs owned by a particular AS. We then used the AS to name and country dataset compiled by to map each AS to a country. This allowed us to sum the number of ip's we expect to belong to a particular country. The results can be seen in Figure 3.

The first observation to make is that multicast makes up a large proportion of Chicago 2008 link's source and destination packets. We did not expect a significant portion of all traffic to be multicast, and we will discuss in detail why we think this happened in section 7. We also note that for each year/location, a few countries make up a large proportion of the traffic, which then falls off quickly for the rest. The US is the top country throughout each dataset, as expected, and China is the second most occurring for 3/4 of the datasets we've analyzed. For the traffic not occupied by multicast and the heavy-hitters, we see that there is a lot of international traffic distributed across the globe. Each of the four we have documented all show unique patterns for high traffic with a particular country. For example, Chicago 2008 sees a lot of German (DE) traffic compared to the rest of the datasets. Chicago 2016 has a large amount of Japanese (JP) traffic, Chicago 2013 sees more

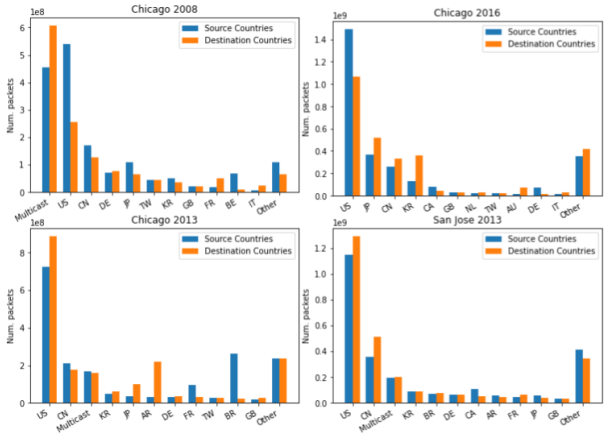


Figure 3: Number of packets based on country origin observed per dataset

Brazilian traffic than the other datsets, and Korea sees a lot of traffic in San Jose link.

5.1.1 Different years. The clearest trend we can see is that as time increased, the amount of multicast seen decreased. In 2008, we saw 28% and 44% of source and destination traffic respectively were designated for multicast, whereas in 2016 that amount went to less than 1% for both. We will not make any assumptions on whether multicast traffic was international or domestic, and so the rest of our discussion will exclude multicast from considerations.

With this assumption in mind, we see in Figure 4 that for all times except for 2016 source IP addresses, there is a larger proportion of

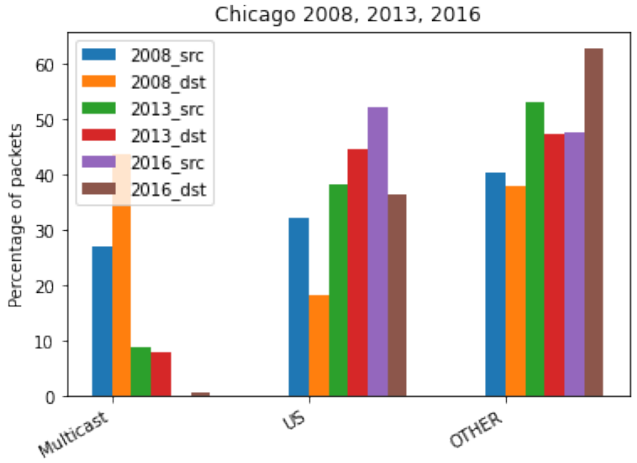


Figure 4: International vs. domestic traffic over time

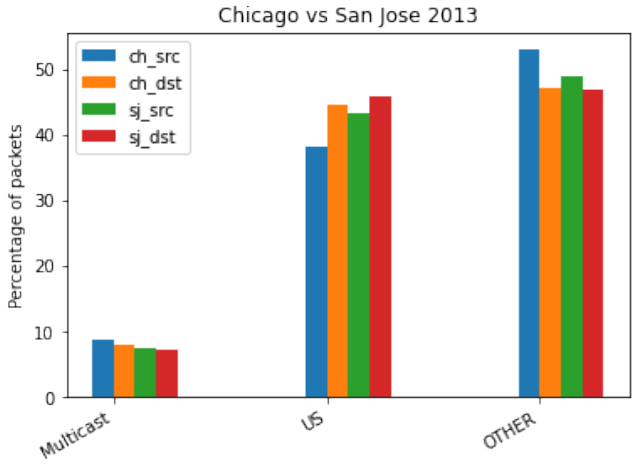


Figure 5: International vs. domestic traffic between cities

international traffic than domestic. The largest of these differences is 2016 destination IP addresses, which had 26% more international traffic than domestic, and 2008 destination IP addresses, which had a 19% difference. This, however, does not indicate a clear upward trend in international traffic over time, as 2013's destination IP addresses only had a 3% difference between domestic and international traffic. The rest of the data we have are within 10% for international vs. domestic traffic. For source IP addresses, we once again do not have clear evidence of an upward trend for domestic or international traffic. We do not feel like this data suffices to make a concrete statement on whether our hypothesis was right in that the proportion of international vs. domestic traffic would stay the same, as the sample size of 3 years may be too small to fully determine this.

5.1.2 Different cities. As seen in Figure 5, the San Jose and Chicago links are more similar than we initially expected. The percentage of multicast traffic for San Jose and Chicago are within 2% of each other. Once again, we will not make any assumptions on whether multicast traffic was international and domestic.

We see slightly more international traffic from Chicago source addresses than San Jose source addresses by 5%, but the percentage of international traffic is within 1% for destination addresses. This is in stark contrast from our assumption that San Jose would have a larger amount of international traffic. There are many factors that may have played into this. One may be that the Chicago to Seattle link helps connect the West Coast to Central and Eastern North America, and so a lot of international traffic designated to those areas could go through this link. Another likely reasoning could be the issues with anonymized IP addresses which we will discuss in a section 7.

6 ADOPTION OF SECURE PROTOCOLS

As the Internet becomes more widely used, more and more information are being stored and retrieve through the Internet, for various services from social media to official government paperwork. Because of this, the importance of encrypting traffic becomes more important, to prevent attackers from reading traffic data, and potentially exposing private information, such as passwords and personal information.

Specifically, we analyze usage of protocols for application layer communication, domain lookup, and email over time. We picked these protocols since they are foundational to much Internet usage: HTTP is the fundamental protocol for the majority of Web usage; DNS is used to translate human-readable website addresses to computer-readable IP addresses; and email is a very common method of communication between users. HTTP is served on port 80, DNS on port 53, and email uses SMTP on port 15 for sending, and IMAP on port 143 or POP on port 110 for receiving. However, these protocols are not encrypted, and secure versions of them have been developed since. HTTPS is the standard protocol for encrypted application layer communication, served on port 443. The standard secure domain lookup protocol is DNS over TLS (DoT), standardized in 2016, and served on port 853. An alternative to DoT is DNS over HTTPS (DoH), also served on 443 like HTTPS, but it is more recent, being standardized in 2018, and is not enabled by default on some browsers, according to Cloudflare [1]. We will look at DoT vs. DNS, since it has a dedicated port, which eases our analysis. SMTP, IMAP, and POP all have secure versions over TLS or SSL, served on ports 587, 993, and 995 respectively. IANA also used to designate port 465 for secure SMTP over SSL, however this has been revoked since, in favor of SMTP over TLS. It should be noted that some SMTP servers on port 25 can also support TLS as well. We will look at all 4 ports for secure email vs. all 3 ports for traditional email.

Research has been done on the adoption of secure protocols for these applications. For application layer communication, Felt et. al. measured the adoption of HTTPS on the web by surveying browsing metrics from major browsers (Google Chrome and Mozilla Firefox), as well as HTTPS support in popular websites, and found substantial HTTPS adoption growth over time [3]. As for DNS, Doan et. al. has shown that only 0.4% of local resolvers from 3200 RIPE Atlas probes are DoT-capable, meaning adoption of DoT is still low [2]. García et. al. also showed that while DoT requests

Table 4: HTTPS vs. HTTP traffic (in num. packets) across prefixes who uses HTTP in all 3 years

	From port 443	From port 80	Total
CHI 08	17,195,598	414,743,680	431,939,278
CHI 13	97,834,854	470,040,372	567,875,226
CHI 16	214,371,650	157,536,021	371,907,671

Table 5: Total HTTPS vs. HTTP traffic (in num. packets)

	From port 443	From port 80	Total
CHI 08	26,831,992	919,659,450	946,491,442
CHI 13	168,877,900	1,265,985,050	1,434,862,950
CHI 16	620,265,804	731,955,947	1,352,221,751

and flows are generally higher than that of DoH, normal DNS still has more usage. We hypothesize from these observations that we would see increased adoption of HTTPS over time compared to HTTP, but DoT adoption will still be low compared to DNS [8]. As for emails, it is reported that port 25 is now blocked by ISPs [13]. As such, we hypothesize that usage of secure email protocols will increase over time as well.

Since we want to measure adoption of protocols, we want to look at the same set of prefixes across the years. For HTTPS vs. HTTP, we look at the Chicago datasets for 2008, 2013, and 2016, where we combine traffic from both directions. We first find the common prefixes in all 3 years that saw any traffic from or to port 80. Then, for each year, we count the total number of packets from or to port 443 and the total number of packets from or to port 80. The results are presented in Table 4. We also counted the total number of HTTPS and HTTP packets across all prefixes between the years in Table 5.

We see that over time, for the same group of prefixes that use HTTP across all 3 years, the number of HTTPS packets increases. The number of HTTP packets, on the other hand, increases from 2008 to 2013, but then decreases from 2013 to 2016. We see that HTTPS is indeed being adopted over time. As for HTTP traffic, we think that either applications that use HTTP to transport information have decreased usage, or they are being routed on a different link. If we look at the total result, we also see the same trend that HTTPS usage is increasing across the years, whereas HTTP traffic fluctuates. We also observe that the amount of “new" traffic (traffic not common between all 3 years) is surprisingly not drastically higher than traffic common bettwen all years. For example, in 2008, over half of HTTPS traffic comes from the common prefixes, and in 2016, HTTPS traffic in the common prefixes takes around a third of all HTTPS traffic. We think this means that this link retains a relatively large amount of hosts that are source of HTTPS traffic.

For DoT vs. DNS, since the secure protocols are launched in mid-2016, only our New York City dataset can be used to measure adoption. We noted in section 3 that we only processed 83% of the 2018 New York dataset, so we present our results with that in mind. We take the same approach as for HTTPS and HTTP, for ports 853 vs. 53. Results are presented in Table 6 and Table 5.

We see that DoT traffic is still generally very miniscule compared to normal DNS. Even though there’s an increase from 2018 to 2019,

Table 6: DoT vs. DNS traffic (in num. packets) across prefixes who uses DNS in all 2 years

	From port 853	From port 53	Total
NYC 18	234	22,708,393	22,708,627
NYC 19	1,064	18,711,418	18,712,482

Table 7: Total DoT vs. DNS traffic (in num. packets)

	From port 853	From port 53	Total
NYC 18	251	35,180,486	35,180,737
NYC 19	1,616	48,420,666	48,422,282

Table 8: Total secure email vs. unsecure email traffic (in num. packets) across prefixes who use SMTP in all 3 years

	To 465/587 From 993/995	To 25 From 110/143	Total
CHI 08	2,136,732	18,878,685	21,015,417
CHI 13	107,637	6,878,909	6,986,546
CHI 16	290,828	1,104,182	1,395,010

both within common prefixes and overall, the actual number is still several orders of magnitude lower than that of DNS traffic. This aligns with the research result seen from more recent papers, detailed earlier in this section, where the amount of traffic using DoT is still mostly lower than that of DNS, presumably because if nowadays DoT-compatible servers are still rare, it would be even rarer in 2018 and 2019. We also see that, as with HTTPS, there is a very large overlap of packets between common prefixes and overall. For both 2018 and 2019, the majority of DoT packets are from prefixes common between two years, whereas for regular DNS, the difference increases from 2018 to 2019. We think that this is because of the lack of available DoT servers local to users. So with an increase in users, there is not necessarily an increase in computers close to DoT servers.

We also see that for the incomplete 2018 dataset, even though we only have 83% of the data, we still see more DNS packets in the common dataset than in 2019. If we had the remaining data, we would expect to see an increase in the number of packets, since we might see more prefixes and more packets with a port 853 or 53. Regardless, given what we’ve seen for 83% of 2018, we doubt that there will be a drastic enough change that leads to different conclusions.

Lastly, for emails, we use the same method as the previous two protocols. We look at the Chicago datasets again. Since we look at two kinds of ports, we look at packets to ports 587 and 465 and packets from ports 993 and 995. This is because clients would send mail to ports 587 and 465, and read emails from ports 993 and 995. We compare that to packets to port 25 and from ports 110 and 143, for the same reasons.

We note that there is a drop in secure email protocol usage from 2008 to 2013, but then an increase from 2013 to 2016. On the other hand, unsecure email protocol usage, and general email usage itself, dropped between all years. We theorize that this is

because throughout the years, as ISPs start preventing usage of unsecure email protocols, these packets would get dropped, or the email clients that use these protocols would be updated to use secure protocols. Despite the increase in number of packets that use secure protocols, the majority still uses the unsecure protocols. We think that this is because there are some ISPs that still don't block these packets. It is also likely that since port 25 communication specifically still supports TLS (via StartTLS), mail clients can still use them.

We also think that the decrease in usage of unsecure protocols, as well as the overall decrease in usage of email through these ports, is because users also have the option to use email clients on the browser, such as the Gmail web app. These traffic use HTTP/S instead of traditional email ports, and we would not see which HTTP traffic is email. Also, over time, there are more alternative methods of communication, such as instant messaging services (iMessage, Facebook Messenger, WhatsApp) for personal communication, and communication platforms (Slack, Discord) for group and organization communication. Because of this, we think that seeing a decrease in mail usage makes sense, as now there are faster and more convenient methods of communication.

7 REFLECTIONS

In this section, we will discuss what we have learned throughout this project, the pitfalls we ran into, and how we can improve for future projects. We have both gained valuable skills regarding finding interesting projects to work on, reading through similar papers for ideas and methodologies, performing large-scale data aggregation from a tcpdump, and answering interesting questions to answer given data.

Now, we would like to discuss an issue with our methodology for collecting data on the IP packets. Our became suspicious as we discovered that a large proportion of traffic in our links for the earlier years was multicast for both sources and destinations, which doesn't make sense, as multicast is not typically used for sources. This led us to analyze the encryption methods in the dataset more closely.

We found that IP addresses in all packets in the anonymized Internet traces dataset are anonymized using the Cryptography-based Prefix-preserving Anonymization (Crypto-PAn) algorithm, which anonymizes the IP addresses themselves, while preserving the prefix relationships between them [16]. Specifically, if two original IP addresses share the first k bits $a_1 a_2 \dots a_k$, then their respective anonymized IP addresses also share the first k bits $a'_1 a'_2 \dots a'_k$. However, the new shared prefix might be different from the original shared prefix.

This means that we should not have directly used IP addresses and prefixes from the CAIDA pfx2as dataset in our analysis, since the IP addresses in the trace are anonymized, and do not fully reflect the actual IP addresses and prefixes that traversed the backbone link. Because we mapped anonymized IP addresses to prefixes that did not undergo anonymization, we were left with inaccurate information regarding the mappings from IP addresses to their associated AS. Thus, the analysis on geographic locations of IP addresses is flawed in this way. With the secure protocol studies, however, we note that while we did in involve prefixes to compare adoption of secure protocols over time, we do not expect this to have a large effect on the analysis and data. This is because IP's within similar prefixes in the anonymized address space will still be mapped to similar prefixes in the real address space. Thus, we would not expect that particular experiment to change much given the real IP addresses of our data. In addition, for our secure protocol adoption analysis, since we relied on prefixes to find common sources across years to measure adoption. This means that the real-world prefixes we would have used might have been different, and we might see different patterns in the results, compared to what we have.

We would like to discuss the implications of this incorrect methodology. Generally, when looking at IP addresses, we would split them into those with or without an identified prefix. We generally see anywhere between 20% to 50% addresses without an unidentified prefix. With a correct address space, we would expect this number to go down, as real-world IP addresses should match up more nicely with real-world prefixes. There will still be some IP addresses that don't have a prefix, since there might be "dummy" packets that are used for testing, but we expect this number to be low, and not publicly traversing the Internet. As for the prefix distribution itself, we think that over time, we would see more fragmentation, as the CAIDA pfx2as generally have an increasing number of prefixes over time. We expect that our results will shift towards a more balanced set of prefixes. There are two cases: some addresses will map to a more specific prefix (for example, from /16 to /24), and some to a broader prefix (for example, from /16 to /8). We expect to see more of the former case, since many addresses got classified to multicast, which is a very broad range (/4). We cannot exactly predict what would happen to geological distribution and secure protocol adoption, but we would expect some changes in those aspects as well, such as different source/destination countries, and different patterns of port usage.

If we have access to a corresponding pfx2as table with the given keys, we believe that obtaining accurate result would be trivial, since we set up our analysis pipeline to be able to take in various pieces of the dataset. Regardless, from this experience, we learned an important lesson in fully understanding how your data is organized, encrypted, and represented before performing any data aggregation or analysis. While the CAIDA website states that it can be used for "geographic and topological distribution," we should have done more research into how this could be done with the limitations of the data. Moving forward, this will be kept at the forefront of our mind when starting a project.

8 CONCLUSION

In this paper we provide insight into understanding Internet traffic trends based upon the CAIDA anonymized Internet traces dataset. We extracted data from the years 2008 and 2013 for the provided San Jose link, 2008, 2013, and 2016 for the Chicago link, and 2019 for the New York link. For each of these, we store information about each packet including the source and destination IP's along with their ports, and the associated prefixes with each. With each of these, we also gave baseline statistics, such as the number of IPv4 and IPv6 packets. We then compared the amount of foreign and domestic traffic from our IP set and found no insights into the relationships on location of the links and their amount of foreign traffic, nor did

we find a large increase or decrease over time. We did, however, see a large decrease in the amount of multicast traffic from 2008 to 2016, where 2016 we had less than 1% of all traffic being multicast. Our study provided us real-world exposure Internet traffic and data analysis, and allowed us to think critically about trends we could find throughout the data we collected, and ideas for how to improve our methods moving forward.

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